

Valence-Arousal Ground Truth Construction for Emotion Geometry Analysis

Deep methodological note for a NeurIPS-style submission: citation chain, numerical reference table, defensible assumptions, robustness checks, and paper-facing wording.

Purpose. This document formalises how discrete emotion labels in the NeuroEmo/LLM emotion-geometry project can be mapped to continuous valence-arousal (V-A) coordinates for global representational analysis. It is written to be inserted into a paper supplement, Methods appendix, or internal reproducibility package. The central aim is to make the V-A mapping transparent, defensible, and reviewer-safe.

1. Executive summary

The project compares geometric structure in emotion representations across language-model embeddings and fMRI-derived brain representations. Some analyses require a continuous affective reference space so that low-dimensional neural or embedding axes can be interpreted as valence-like or arousal-like. Because the source fMRI dataset supplies categorical labels rather than participant-level V-A ratings, we construct an explicit reference table that maps each discrete label to approximate coordinates in a standard affective space.

This mapping should be presented as a **reference coordinate system**, not as directly measured subjective ground truth. The correct claim is not that the dataset measured valence and arousal for every participant. The correct claim is that the categorical labels are positioned within a widely used affective framework so that representational axes can be compared with established affective structure.

The citation chain has two levels. First, Russell (1980), building on the broader circumplex tradition including Mehrabian and Russell (1974), provides the theoretical model: affective states can be organised by dimensions corresponding approximately to valence/pleasantness and arousal/activation. Second, empirical rating resources such as DEAP (Koelstra et al., 2012), ANEW (Bradley and Lang, 1999), and IAPS (Lang, Bradley, and Cuthbert, 2008) provide numerical affective ratings on valence-arousal scales. The project should cite Russell for the coordinate framework and cite empirical norm datasets for the use of numerical ratings.

2. Why a valence-arousal mapping is needed

The primary analysis uses categorical emotion labels such as Afraid, Calm, Delighted, Depressed, and Excited. These labels are sufficient for classification, centroid construction, overlap analysis, and prototype-margin calculations. However, they are not sufficient for interpreting whether a global representational axis corresponds to a known psychological dimension. For example, a principal component of brain emotion centroids may separate high-activation emotions from low-activation emotions. To test whether that component is arousal-like, each emotion category must be associated with a numerical arousal coordinate.

The purpose of the V-A mapping is therefore interpretive and comparative. It allows the paper to ask questions such as: Does the first neural centroid axis align more strongly with arousal than valence? Do LLM embedding axes show the opposite pattern? Is global geometry different even when local ambiguity/margin structure is shared? These are higher-level representational questions, not claims about each subject's reported feeling during scanning.

This distinction should be explicit in the manuscript. The assigned coordinates are **external reference values** derived from affective science, not measured responses from the NeuroEmo participants. The mapping should be used to support global-axis interpretation, while the main ambiguity claim should remain based on directly observed geometry and classifier/logit behaviour.

3. Citation chain and source roles

Source	Role in the paper	What it supports	What it does not support
Russell (1980), A circumplex model of affect	Primary theoretical foundation	Affective states can be organised in a circular space structured by dimensions closely corresponding to pleasantness/valence and activation/arousal.	Exact numerical coordinates for this project's five emotion labels.
Mehrabian and Russell (1974), An approach to environmental psychology	Precursor dimensional affect framework	The broader tradition of affective dimensions involving pleasure, arousal, and dominance.	Dataset-specific values for Afraid/Calm/Delighted/Depressed/Excited.
Koelstra et al. (2012), DEAP	Empirical numerical anchor	Continuous valence and arousal ratings for affective stimuli, rated by participants after viewing music videos.	Direct category centroids for every label used in NeuroEmo. DEAP is stimulus-rating data, not a direct label dictionary.
Bradley and Lang (1999), ANEW	Lexical affective norms	Valence, arousal, and dominance ratings for English emotion-related words, useful for anchoring categorical labels as word concepts.	Brain-state measurement or fMRI participant ratings.
Lang, Bradley, and Cuthbert (2008), IAPS	Visual affective norms	Normative ratings of affective pictures, supporting the common use of V-A coordinates for visual/emotional stimulus interpretation.	Exact values for every project label unless matching stimuli are used.
Posner, Russell, and Peterson (2005)	Neuroscience-facing review	Circumplex interpretation in affective neuroscience; supports relevance of V-A dimensions to neural emotion processing.	Primary numerical ratings.

4. Recommended reference table

The table below provides a defensible working coordinate system for the five primary brain labels and the sixth LLM label often used in the embedding experiments. Values are placed on a 1-9 Self-Assessment Manikin-style scale, where higher valence means more pleasant and higher arousal means more activated. The values should be treated as approximate category-level coordinates. If a final submission uses different exact numbers, the same principles apply: keep the table explicit, cite the source chain, and run sensitivity checks.

Emotion label	Recommended valence	Recommended arousal	Circumplex interpretation	Rationale
Afraid / Fear	2.5	7.5	negative, high arousal	Fear is consistently positioned as unpleasant and highly activated because it reflects threat, vigilance, and defensive mobilisation.
Calm	7.0	2.5	positive, low arousal	Calm is pleasant or mildly positive but deactivated; it is the opposite quadrant from fear/excitement on arousal.
Delighted / Happiness	8.2	6.5	positive, moderate-high arousal	Delight is strongly pleasant and activated; it overlaps conceptually with joy and high-positive affect.
Depressed / Sadness	2.2	3.0	negative, low arousal	Depressed affect is unpleasant and low in activation; it sits in the low-valence, low-arousal quadrant.
Excited	7.8	8.2	positive, high arousal	Excitement is pleasant and highly activated, often more arousing than generic happiness.

Emotion label	Recommended valence	Recommended arousal	Circumplex interpretation	Rationale
Love	8.4	5.5	positive, moderate arousal	Love is strongly pleasant but often less uniformly high-arousal than excitement; exact arousal depends on context.

Important wording rule. Do not write that these values are the measured ground truth of the participants. Write that they are a predefined external reference mapping based on the circumplex model and empirical affective-norm resources.

5. Why these coordinates are scientifically defensible

5.1 Theoretical defensibility

The circumplex model treats affective states as positions in a two-dimensional space rather than as unrelated discrete boxes. The two most commonly used axes are valence and arousal. This is exactly the structure needed for the project: it permits categories such as Afraid, Calm, Delighted, Depressed, and Excited to be compared by their relative locations rather than by arbitrary label identity.

The chosen coordinates preserve the quadrant structure predicted by the circumplex. Afraid is low-valence/high-arousal; Calm is high-valence/low-arousal; Delighted and Excited are high-valence/high-arousal; Depressed is low-valence/low-arousal. This quadrant structure is more important than the exact decimal values, because the representational analysis primarily tests whether neural or embedding axes align directionally with valence or arousal.

5.2 Empirical defensibility

DEAP, ANEW, and IAPS demonstrate that affective stimuli and affective words can be rated reliably on continuous valence and arousal dimensions. DEAP is especially useful because participants rated music videos after exposure, making it relevant for stimulus-evoked affect rather than purely lexical concepts. ANEW is useful because project labels are words, and it provides normative word-level ratings. IAPS is useful because it shows the same rating framework applied to visual emotional stimuli.

No single source should be portrayed as providing an exact lookup table for all project labels. Instead, the paper should state that the mapping is based on convergent affective-science norms. This is more honest and harder for reviewers to attack.

5.3 Analytical defensibility

The V-A table is not the main evidence for the paper. It is used to interpret global axes after the representational structure has already been estimated. The central ambiguity result is based on centroid margin, classifier uncertainty, and logit behaviour. The V-A mapping plays a narrower role: it helps explain whether the global ordering of emotion centroids is arousal-dominant, valence-dominant, or neither.

6. Mathematical use in the analysis

Let each emotion category k have an assigned external coordinate v_k for valence and a_k for arousal. Let p_{k1} and p_{k2} be the first two coordinates of the projected emotion centroid, for example the first two principal component scores of the five emotion centroids. The alignment analysis can then correlate each representational axis with the external V-A coordinates.

For each axis m :

$$\begin{aligned}\text{corr}(\text{PC}_m, \text{Valence}) &= \text{corr}(\{p_{km}\}_{k=1..K}, \{v_k\}_{k=1..K}) \\ \text{corr}(\text{PC}_m, \text{Arousal}) &= \text{corr}(\{p_{km}\}_{k=1..K}, \{a_k\}_{k=1..K})\end{aligned}$$

Because K is small for the brain dataset, correlations should be interpreted cautiously. Report effect sizes, p -values, and ideally sensitivity analysis. The safest statement is not that the brain universally organises emotion by arousal, but that the dominant centroid axis in this dataset aligns more strongly with the assigned arousal reference than with the assigned valence reference.

7. Sensitivity analysis plan

A reviewer may object that the numerical V-A values are approximate. The correct response is to show that the qualitative result is robust to reasonable perturbations of those values. This converts a vulnerability into a strength.

Recommended procedure:

Step	Operation	Purpose
1	Define the baseline V-A table in the appendix.	Makes assumptions transparent and reproducible.
2	Add random Gaussian noise to valence and arousal values, e.g. SD = 0.25, 0.50, and 0.75 on the 1-9 scale.	Tests whether conclusions depend on exact decimals.
3	Repeat the PC-axis correlation analysis 1,000-10,000 times.	Produces a robustness distribution.
4	Report the proportion of perturbations where the same qualitative result holds, e.g. PC1 remains more arousal-aligned than valence-aligned.	Shows the finding reflects quadrant structure, not fragile point estimates.
5	Optionally repeat using rank correlations.	Protects against nonlinear scaling and exact-value concerns.

Suggested robustness sentence: *Because the V-A coordinates are category-level approximations rather than participant-specific measurements, we repeated the alignment analysis under random perturbations of the assigned coordinates. The qualitative arousal-dominant alignment remained stable across perturbations, indicating that the result is not an artefact of a particular numerical mapping.*

8. Paper-facing Methods text

The following paragraph is written to be copied into the Methods section of a NeurIPS-style manuscript.

Valence-arousal reference mapping. To interpret global representational axes, we assigned each discrete emotion label an external coordinate in valence-arousal space. This mapping was grounded in the circumplex model of affect, which organises affective states along pleasantness/valence and activation/arousal dimensions (Russell, 1980; Mehrabian and Russell, 1974). Numerical coordinates were informed by empirical affective-norm resources including DEAP, in which participants provide continuous valence and arousal ratings for emotion-eliciting videos (Koelstra et al., 2012), and lexical/visual affective norms such as ANEW and IAPS (Bradley and Lang, 1999; Lang et al., 2008). Because the fMRI dataset contains categorical emotion labels rather than participant-level V-A ratings, these coordinates were used as an external reference geometry rather than as measured subjective ground truth. We therefore interpret V-A alignment as a comparison between representational axes and established affective structure, not as direct evidence of individual affective experience.

9. Paper-facing Results text

The following paragraph is written to be copied into the Results section, with numerical values replaced by the final verified outputs.

Global neural geometry is more arousal-aligned than valence-aligned. After projecting emotion centroids into the leading two-dimensional representational subspace, we correlated each centroid axis with the external V-A reference coordinates. The first neural axis showed substantially stronger alignment with arousal than with valence (PC1-arousal: $r = [\text{INSERT}]$, $p = [\text{INSERT}]$; PC1-valence: $r = [\text{INSERT}]$, $p = [\text{INSERT}]$). The second axis did not show comparable alignment with either dimension. This suggests that, in the fMRI-derived ROI representation, the dominant global ordering of emotion centroids is more strongly related to activation/arousal than pleasantness/valence. Because only five emotion centroids are available, we treat this result as an interpretive global-structure analysis rather than as the primary evidence for the paper. The central ambiguity result remains the margin-uncertainty relationship.

10. Paper-facing Limitations text

The following paragraph should appear in the Limitations or Appendix.

The V-A analysis relies on externally assigned category-level coordinates rather than participant-specific ratings collected during the NeuroEmo experiment. This is necessary because the source dataset provides discrete labels but not continuous subjective affect ratings for each subject and condition. The assigned coordinates are therefore approximate and should not be interpreted as measured ground truth. To reduce this limitation, we make the mapping explicit, cite both theoretical and empirical sources, and recommend sensitivity analyses under reasonable perturbations of the coordinates. The global-axis findings should be interpreted as supportive evidence about representational organisation, not as the main proof of the margin-based ambiguity mechanism.

11. Exported file recommendation

For reproducibility, the project should export a dedicated CSV file that contains exactly the values used in the paper. Recommended filename:

valence_arousal_reference_table.csv

Recommended columns:

Column	Description
emotion	Emotion label exactly as used in the analysis.
valence_1_9	Assigned valence coordinate on a 1-9 scale.
arousal_1_9	Assigned arousal coordinate on a 1-9 scale.
quadrant	Circumplex quadrant label, e.g. low-valence/high-arousal.
primary_theory_source	Usually Russell 1980 / Mehrabian and Russell 1974.
numeric_anchor_source	DEAP, ANEW, IAPS, or project-specific affective-norm mapping.
notes	Short justification and caveat.

12. Example CSV content

emotion	valence_1_9	arousal_1_9	quadrant	notes
Afraid	2.5	7.5	low-valence/high-arousal	Threat-related negative activation; fear-like quadrant.
Calm	7.0	2.5	high-valence/low-arousal	Pleasant deactivation; relaxation/calm quadrant.
Delighted	8.2	6.5	high-valence/high-arousal	Joy/delight; high positive affect with activation.
Depressed	2.2	3.0	low-valence/low-arousal	Sad/depressed low activation quadrant.
Excited	7.8	8.2	high-valence/high-arousal	Highly activated positive affect.
Love	8.4	5.5	high-valence/moderate-arousal	Strongly pleasant, context-dependent arousal.

13. Reviewer-risk audit

Potential reviewer criticism	Why it matters	How this document handles it
The values look arbitrary.	Undermines credibility of VA alignment.	The table is explicit, theoretically grounded, and tied to empirical affective norm datasets.
Russell does not provide exact values.	Citation mismatch.	Russell is cited for the coordinate model; DEAP/ANEW/IAPS are cited for numerical affective rating practice.
The dataset did not measure valence/arousal.	Could be seen as overclaiming.	The text repeatedly states the mapping is external reference geometry, not participant-level ground truth.

Potential reviewer criticism	Why it matters	How this document handles it
Only five brain emotion categories.	Small K makes correlations fragile.	The Results text treats VA alignment as supportive, recommends sensitivity checks, and keeps margin as the primary result.
Exact decimals may alter correlations.	Can weaken quantitative claims.	The sensitivity plan perturbs coordinates and tests qualitative stability.

14. References

Bradley, M. M., and Lang, P. J. (1999). Affective Norms for English Words (ANEW): Instruction Manual and Affective Ratings. Technical Report C-1, The Center for Research in Psychophysiology, University of Florida.

Koelstra, S., Muhl, C., Soleymani, M., Lee, J. S., Yazdani, A., Ebrahimi, T., Pun, T., Nijholt, A., and Patras, I. (2012). DEAP: A Database for Emotion Analysis Using Physiological Signals. *IEEE Transactions on Affective Computing*, 3(1), 18-31. doi:10.1109/T-AFFC.2011.15.

Lang, P. J., Bradley, M. M., and Cuthbert, B. N. (2008). International Affective Picture System (IAPS): Affective Ratings of Pictures and Instruction Manual. Technical Report A-8. Gainesville, FL: University of Florida.

Mehrabian, A., and Russell, J. A. (1974). *An Approach to Environmental Psychology*. Cambridge, MA: MIT Press.

Posner, J., Russell, J. A., and Peterson, B. S. (2005). The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology. *Development and Psychopathology*, 17(3), 715-734.

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161-1178.

15. Final recommended wording discipline

Use: "external V-A reference coordinates", "category-level approximation", "global-axis interpretation", "alignment with affective reference structure".

Avoid: "ground-truth feelings", "measured valence and arousal", "the brain proves arousal is the universal organising axis", "exact affective coordinates".

The final defensible claim is: *The dominant global axis of the tested neural emotion geometry aligns more strongly with an externally defined arousal reference than with an externally defined valence reference, while the main ambiguity mechanism is supported independently by centroid margin and classifier/logit uncertainty.*